测定介质中的声速

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1、实验数据

1.1 驻波法，相位法，气体参量法测空气中声速

共振频率：$f = 40.71kHz$

1.1.1 驻波法测声速实验数据表

i（增大间距）	1	2	3	4	5
x/mm	41.610	45.020	49.400	53.830	58.140
V_{pp}/V	20.60	19.00	16.60	15.00	13.00
i	6	7	8	9	10
x/mm	62.450	66.950	71.300	74.980	79.500
V_{pp}/V	12.10	10.90	9.76	9.36	8.88
i（减小间距）	1	2	3	4	5
x/mm	79.400	74.690	71.760	66.490	62.260
V_{pp}/V	8.80	9.36	9.76	10.70	11.80
i	6	7	8	9	10
x/mm	57.870	53.450	49.200	44.760	40.300
V_{pp}/V	12.80	14.20	15.70	18.00	20.00

1.1.2 相位法测声速实验数据表

i（增大间距）	1	2	3	4	5
x/mm	43.940	52.390	60.840	69.280	77.850
i	6	7	8	9	10
x/mm	86.280	94.780	103.270	111.710	120.280
i（减小间距）	1	2	3	4	5
x/mm	120.380	111.670	103.140	94.730	86.110
i	6	7	8	9	10
x/mm	77.750	69.170	60.720	52.260	43.790

1.1.3 气体参量法测声速实验数据表

气压$p/mmHg$	温度θ/K	相对湿度H
756.70	19.0	56%

1.2 声光效应法测水中声速

超声波频率：$f = 9.569MHz$　　水温：$T = 22.6℃$

氦氖激光器波长：$\lambda = 632.8nm$

超声光栅到墙壁距离：$L = 461.80cm$

1.2.1 声光效应法测水中声速实验数据表

i（条纹级数）	-5	-4	-3	-2	-1	0
x/mm	-94.0	-75.5	-57.0	-38.0	-19.0	0
i（条纹级数）	1	2	3	4	5	
x/mm	19.0	37.5	56.5	75.5	94.0	

2、数据的分析、处理和结论

2.1 用最小二乘法处理驻波法与相位法的数据

2.1.1 驻波法：增大间距组：

$$k_1 = (4.266 \pm 0.033)mm\left(\sigma_{Ak1} = k_1\sqrt{\frac{1-r^2}{r^2(n-2)}} = 0.032mm\ e = 0.01mm\ \sigma_{Bk1} = \frac{e}{\sqrt{3}} = 0.006mm\right)$$

$r = 0.99978$

减小间距组：

$$k_2 = (4.35 \pm 0.04)mm\ \left(\sigma_{Ak2} = 0.04mm\ e = 0.01mm\ \sigma_{Bk2} = \frac{e}{\sqrt{3}} = 0.006mm\right)$$

$r = 0.99964$

$$\bar{\lambda} = (k_1 + k_2) = (8.62 \pm 0.10)mm\ (\sigma_{\bar{\lambda}} = \sqrt{\sigma_{k1}^2 + \sigma_{k2}^2} = 0.10mm)$$

$$f = (40.710 \pm 0.006)kHz\ \left(e_f = 0.01kHz\ \ \sigma_{Bf} = \frac{e_f}{\sqrt{3}} = 0.006kHz\right)$$

$$v = f\bar{\lambda} = (3.51 \pm 0.04) \times 10^2\ m/s\ \ \sigma_v = \sqrt{(\bar{\lambda}\sigma_f)^2 + (f\sigma_{\bar{\lambda}})^2}$$

$$\epsilon = \frac{\sigma_v}{v} = \sqrt{\left(\frac{\sigma_f}{f}\right)^2 + \left(\frac{\sigma_{\bar{\lambda}}}{\bar{\lambda}}\right)^2} = 1.2\%$$

2.1.2 相位法：增大间距组：

$$k_1 = (8.481 \pm 0.007)mm\ \left(\sigma_{Ak1} = 0.004mm\ e = 0.01mm\ \sigma_{Bk1} = \frac{e}{\sqrt{3}} = 0.006mm\right)$$

$r = 0.9999990$

减小间距组：

$$k_2 = (8.499 \pm 0.011)mm\ \left(\sigma_{Ak2} = 0.009mm\ e = 0.01mm\ \sigma_{Bk2} = \frac{e}{\sqrt{3}} = 0.006mm\right)$$

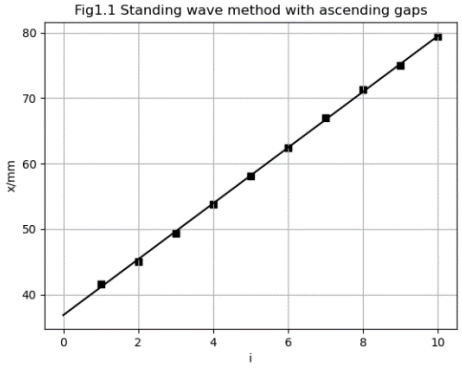
$r = 0.99999953$

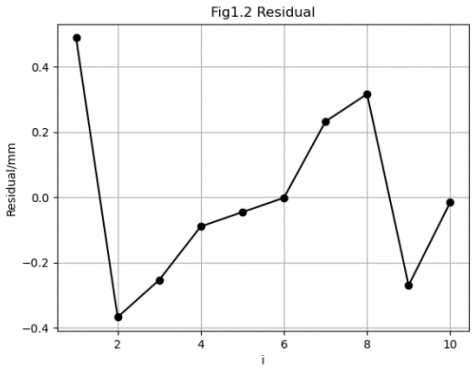
$$\bar{\lambda} = \frac{1}{2}(k_1 + k_2) = (8.490 \pm 0.013)mm\ (\sigma_{\bar{\lambda}} = \sqrt{\sigma_{k1}^2 + \sigma_{k2}^2} = 0.013mm)$$

$$v = f\bar{\lambda} = (3.456 \pm 0.005) \times 10^2\ m/s$$

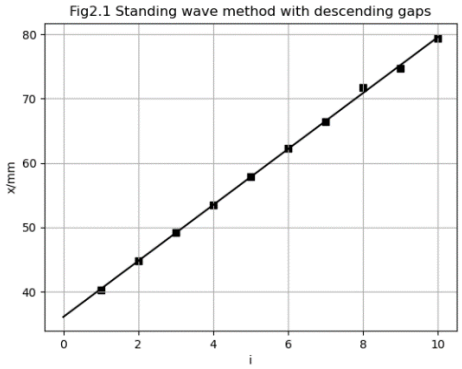
$$\epsilon = \frac{\sigma_v}{v} = 0.15\%$$

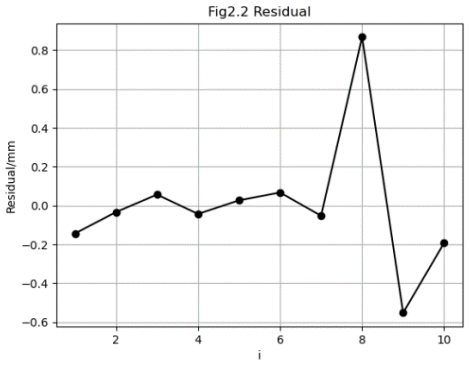
驻波法增大间距组（线性拟合$x \sim i$与残差图）：



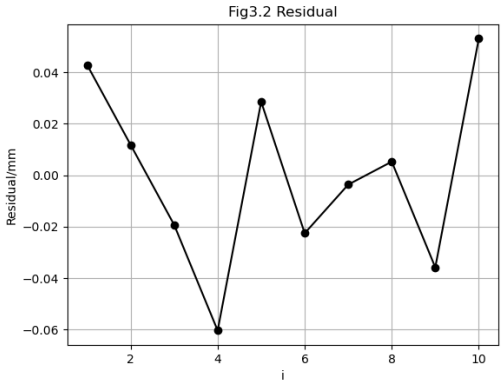


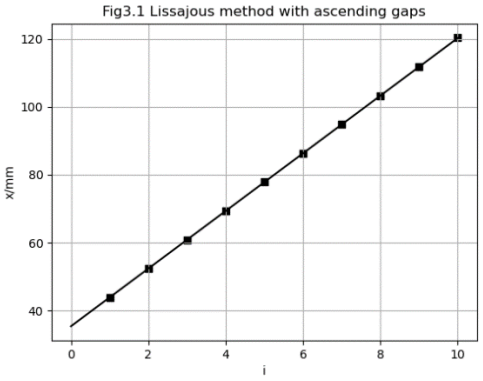
驻波法减小间距组（线性拟合$x\sim i$与残差图）：



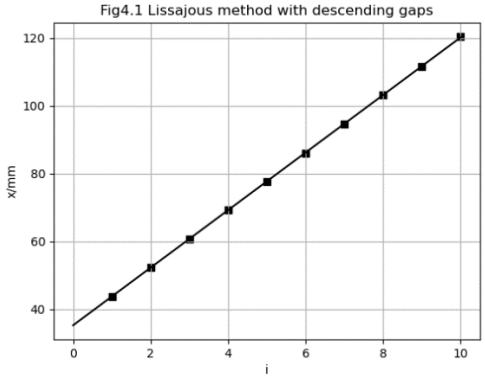


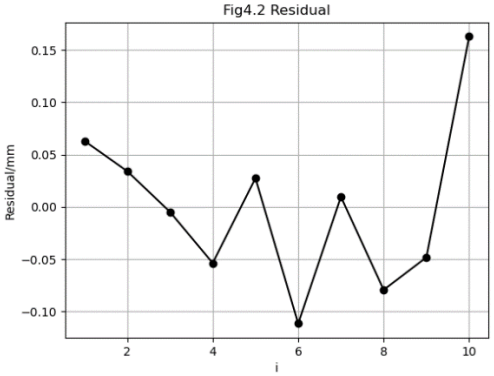
相位法增大间距组（线性拟合$x\sim i$与残差图）：





相位法减小间距组（线性拟合$x\sim i$与残差图）：





2.1.3 分析:

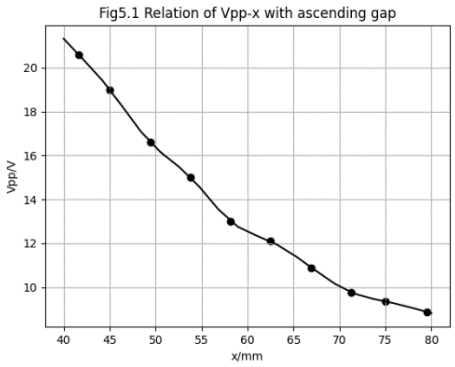
a.由不确定度可知，对于驻波法，增大与减小间距组拟合后斜率的 A 类不确定度比 B 类不确定度大;而对于相位法，拟合后斜率的 A 类不确定度与 B 类不确定度相近。驻波法得出的斜率的主要误差来源于对**电压峰值的判断有距离宽度以及电压峰值可能由于接触等问题引起不稳定**，故用千分尺精度过高；而相位法用李萨如图形会更稳定，位置更精确，用千分尺符合测量要求。同时，相位法得出的相对误差比驻波法得出的小得多，也是因为上述原因。

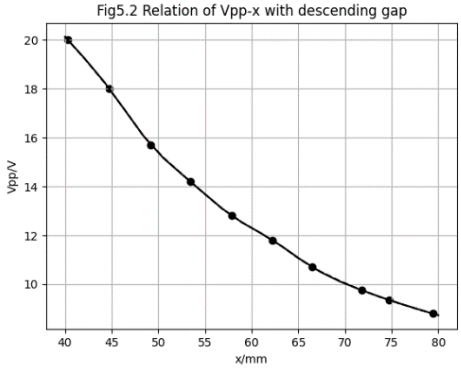
b.对增大与减小间距组数据分别处理后取平均可以减小千分尺的回程差的系统误差

c.驻波法中$\frac{\sigma_{\bar{\lambda}}}{\bar{\lambda}}=1.16\%,\frac{\sigma_f}{f}=0.015\%$，可见主要误差来源于$\bar{\lambda}$；相位法中$\frac{\sigma_{\bar{\lambda}}}{\bar{\lambda}}=0.15\%,\frac{\sigma_f}{f}=0.015\%$，可见主要误差来源于$\bar{\lambda}$

2.2 峰-峰值电压与传播距离的关系

2.2.1 Vpp-x 关系图（三次样条）





2.1.2 分析:

声能密度$I\propto p^2$，**假设** Vpp 近似与声压 p 成正比，因此此处以Vpp^2代表声能密度

如果认为实验中采用线状声源，可用$V_{pp}^2=Ce^{-\alpha x}\left(-lnV_{pp}^2=\alpha x-lnC\right)$去拟合

拟合结果：增大间距组:

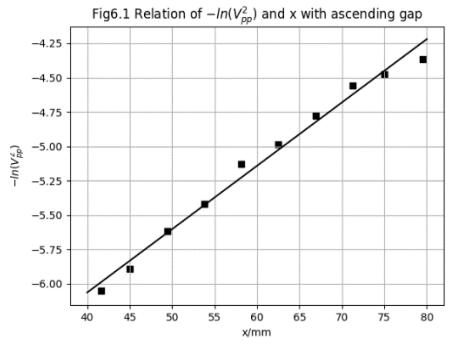
$\alpha=(0.0460\pm0.0018)mm^{-1},lnC=7.90\pm0.11,r=0.9937$

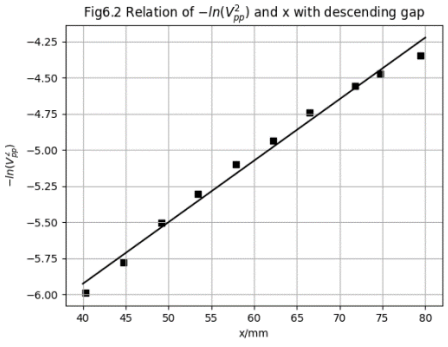
$C=(2.70\pm0.30)\times10^3\ V^2,(\sigma_C=e^{lnC}\sigma_{lnC})$

减小间距组:

$\alpha=(0.0426\pm0.0016)mm^{-1},lnC=7.63\pm0.10,r=0.9942$

$C=(2.06\pm0.21)\times10^3\ V^2$





由此可见，数据与公式符合的相当好

定性结论：在温度为 19 摄氏度，相对湿度为 56%，大气压为1.0079×10^5Pa的条件下，在

一定距离范围内，声波能量随距离的增大而减小，且减小速率随距离的增大而减小

2.3 气体参量法

2.3.1 数据处理

$g = 9.801\ m/s^2$ $\rho_{Hg} = 13.59 \times 10^3 kg/m^3$ $T_0 = 273.15K$

$\theta = (19.0 \pm 0.6)K$ $(e_\theta = 1K, \sigma_\theta = \frac{e_\theta}{\sqrt{3}} = 0.6K)$

查表知该温度下的饱和蒸汽压为$p_s = 2196.9Pa$

$H = (56 \pm 5)\%$ $\left(e_H = 8\%, \sigma_H = \frac{e_H}{\sqrt{3}} = 5\%\right)$

$p = (1.0079 \pm 0.0006) \times 10^5 Pa$ $(e_p = 0.001 \times 10^5 Pa, \sigma_p = \frac{e_p}{\sqrt{3}} = 0.0006 \times 10^5 Pa)$

$p_w = p_s H = (1.23 \pm 0.11) \times 10^3 Pa$ $(\sigma_{pw} = p_s \sigma_H = 0.11 \times 10^3 Pa)$

0 摄氏度时空气中声速为$v_0 = 331.45 m/s$

$v = v_0 \sqrt{\left(1 + \frac{\theta}{T_0}\right)\left(1 + \frac{0.3192 p_w}{p}\right)} = (3.435 \pm 0.007) \times 10^2 m/s$ （4 位有效数字）

$$\sigma_v = \sqrt{(\frac{\partial v}{\partial \theta}\sigma_\theta)^2 + (\frac{\partial v}{\partial p_w}\sigma_{pw})^2 + (\frac{\partial v}{\partial p}\sigma_p)^2} = 0.007 \times 10^2 m/s$$

$$\frac{\sigma_v}{v} = \sqrt{(\frac{\partial lnv}{\partial \theta}\sigma_\theta)^2 + (\frac{\partial lnv}{\partial p_w}\sigma_{pw})^2 + (\frac{\partial lnv}{\partial p}\sigma_p)^2} = 0.20\%$$

2.3.2 分析

a. $\frac{\sigma_\theta}{\theta} = 3.2\%$ $\frac{\sigma_p}{p} = 0.06\%$ $\frac{\sigma_H}{H} = 8.9\%$，这说明主要误差来源于温度与相对湿度的测量

b.驻波法$v_1 = (351 \pm 4) m/s$，相位法$v_2 = (345.6 \pm 0.5) m/s$，气体参量法$v_3 = (343.5 \pm 0.7) m/s$，各相对误差$\frac{\sigma_{v1}}{v_1} = 1.2\%$ $\frac{\sigma_{v2}}{v_2} = 0.15\%$ $\frac{\sigma_{v3}}{v_3} = 0.20\%$，由此可见，驻波法的结果与相对误差明显高于其他两种方法，原因是**电压峰值的判断有距离宽度以及电压峰值可能由于接触等问题引起不稳定**，使得读数与真实的峰值对应的点偏差很大；同时相位法的相对误差比气体参量法的小，原因是气体参量法中要测量的温度与相对湿度误差较大，并且有时不稳定，但相位法主要测量距离，距离测量更方便与稳定，且千分尺精度较高，所以相位法在此实验中相对误差会较小

2.4 声光效应法测水中声速

2.4.1 公式

超声光栅是相位型光栅，光栅常数d为超声波波长λ_s，光栅方程$dsin\theta = k\lambda$，在实验中近似认为θ很小，于是$\lambda_s \frac{\delta x}{L} = \lambda$，$v = f\lambda_s = f\lambda \frac{L}{\delta x}$

2.4.2 用最小二乘法拟合i与x来求出δx

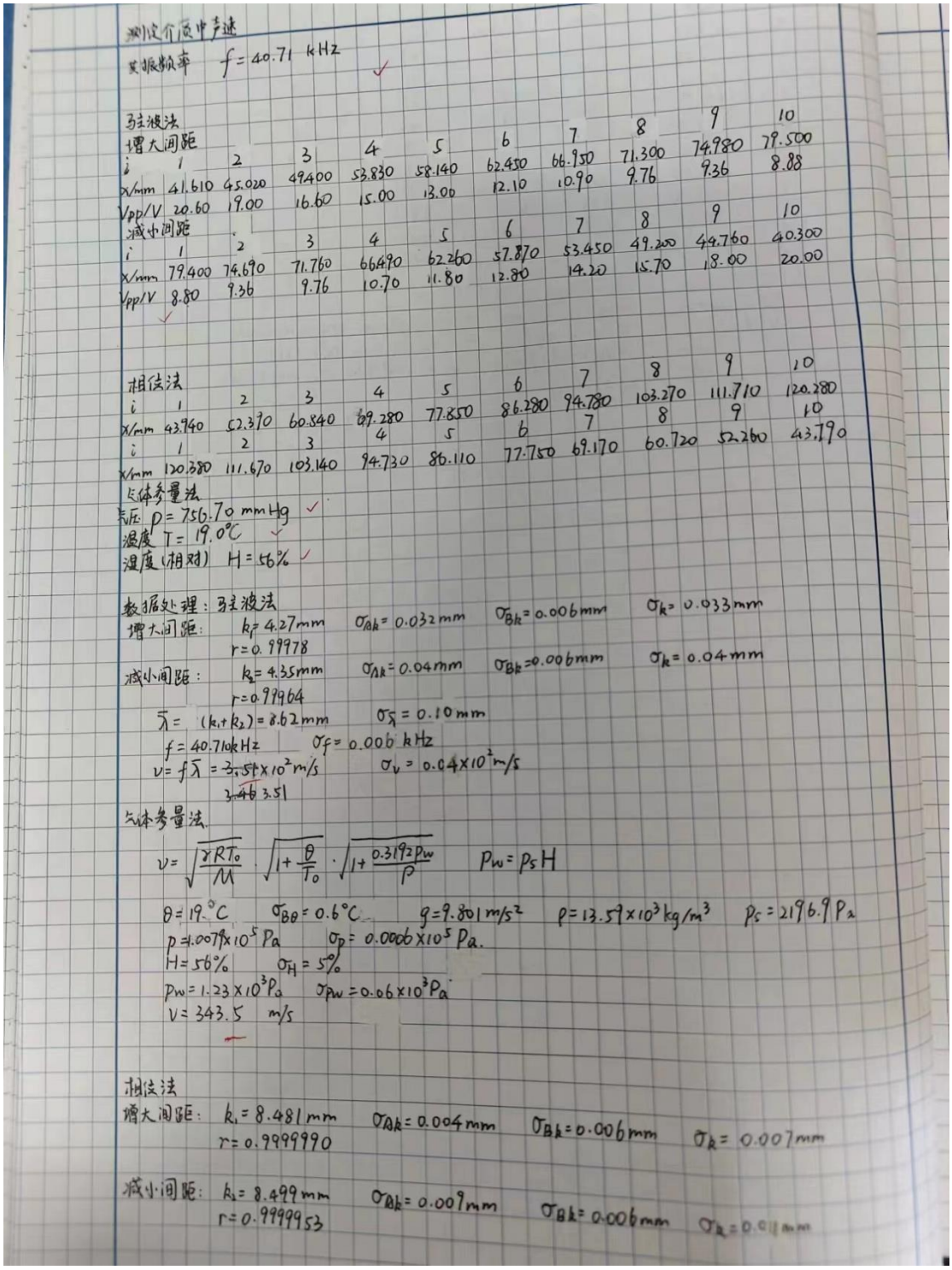
拟合结果：$\delta x = 18.85mm$ $r = 0.999994$

超声波频率：$f = 9.569MHz$

氦氖激光器波长：$\lambda = 632.8nm$

超声光栅到墙壁距离：$L = 461.80cm$

$v = 1483 m/s$



测定介质中声速

发振频率 $f = 40.71$ kHz ✓

驻波法

增大间距

i	1	2	3	4	5	6	7	8	9	10
x/mm	41.610	45.020	49.400	53.830	58.140	62.450	66.950	71.300	74.980	79.500
Vpp/V	20.60	19.00	16.60	15.00	13.00	12.10	10.90	9.76	9.36	8.88

减小间距

i	1	2	3	4	5	6	7	8	9	10
x/mm	79.400	74.690	71.760	66.490	62.260	57.870	53.450	49.200	44.760	40.300
Vpp/V	8.80	9.36	9.76	10.70	11.80	12.80	14.20	15.70	18.00	20.00

✓

相位法

i	1	2	3	4	5	6	7	8	9	10
x/mm	43.940	52.370	60.840	69.280	77.850	86.280	94.780	103.270	111.710	120.280

i	1	2	3	4	5	6	7	8	9	10
x/mm	120.380	111.670	103.140	94.730	86.110	77.750	69.170	60.720	52.260	43.790

气体参量法

气压 $p = 756.70$ mmHg ✓

温度 $T = 19.0°C$ ✓

湿度(相对) $H = 56\%$ ✓

数据处理：驻波法

增大间距： $k_1 = 4.27$mm $\sigma_{Ak} = 0.032$mm $\sigma_{Bk} = 0.006$mm $\sigma_k = 0.033$mm

$r = 0.99978$

减小间距： $k_2 = 4.35$mm $\sigma_{Ak} = 0.04$mm $\sigma_{Bk} = 0.006$mm $\sigma_k = 0.04$mm

$r = 0.99964$

$\overline{\lambda} = (k_1 + k_2) = 8.62$mm $\sigma_{\overline{\lambda}} = 0.10$mm

$f = 40.710$kHz $\sigma_f = 0.006$ kHz

$v = f\overline{\lambda} = 3.51 \times 10^2$ m/s $\sigma_v = 0.04 \times 10^2$ m/s

~~3.46~~ 3.51

气体参量法

$$v = \sqrt{\frac{\gamma RT_0}{M}} \cdot \sqrt{1 + \frac{\theta}{T_0}} \cdot \sqrt{1 + \frac{0.319 p_w}{p}} \qquad p_w = p_s H$$

$\theta = 19.°C$ $\sigma_{B\theta} = 0.6°C$ $g = 9.801\ m/s^2$ $\rho = 13.59 \times 10^3\ kg/m^3$ $p_s = 2196.9\ Pa$

$p = 1.007 \times 10^5$ Pa $\sigma_p = 0.0006 \times 10^5$ Pa.

$H = 56\%$ $\sigma_H = 5\%$

$p_w = 1.23 \times 10^3$ Pa $\sigma_{pw} = 0.06 \times 10^3$ Pa.

$v = 343.5$ m/s

相位法

增大间距： $k_1 = 8.481$mm $\sigma_{Ak} = 0.004$mm $\sigma_{Bk} = 0.006$mm $\sigma_k = 0.007$mm

$r = 0.9999990$

减小间距： $k_2 = 8.499$mm $\sigma_{Ak} = 0.009$mm $\sigma_{Bk} = 0.006$mm $\sigma_k = 0.011$mm

$r = 0.9999953$

$\bar{\lambda} = \frac{1}{2}(k_1+k_2) = 8.49\,mm$ $\sigma_{\bar{\lambda}} = 0.013\,mm$

$f = 40.71\,kHz$ $\sigma_f = 0.005\,kHz$

$v = f\bar{\lambda} = 3.450\times10^2\,m/s$ $\sigma_v = 0.005\times10^2\,m/s$

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声光效应

$f = 9.560\,MHz$ 水温 $22.6^\circ C$

$L' = 463\,cm$ $h = 4.70\,cm$ $L = L' - \frac{h}{2}$

i	-4	-3	-2	-1	0	1	2	3	4
x/mm	-75.0	-58.5	-38.2	-19.4	0.0	19.5	38.5	56.5	75.6

光栅常数 $d = \lambda_s$ $d\sin\theta = k\lambda$ $\sin\theta \approx \frac{\Delta X}{L}$

$\lambda_s \frac{\Delta X}{L} = \lambda$ $\lambda = 632.8\,nm$

拟合后 $\Delta X = 19.00\,mm$ $L = L' - \frac{h}{2} = 460.65\,cm$

$\lambda_s = 1.53\times10^{-4}\,m$

$v = \lambda_s f = 1463\,m/s$

$f = 9.569\,MHz$

$L = 461.3\,cm$ 水温 $22.6^\circ C$

i	-5	-4	-3	-2	-1	0	1	2	3	4	5
x/mm	-94.0	-75.5	-57.0	-38.0	-19.0	0	19.0	37.5	56.5	75.5	94.0

$\Delta X = 18.85\,mm$ $r = 0.999994$

$\lambda_s = 1.55\times10^{-4}\,m$

$v = 1483\,m/s$

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